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Identification and causal analysis of predatory open access journals based on interpretable machine learning

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Abstract
Predatory journals have been a recent phenomenon, drawing attention from the academic community in the last decade. However, as the open access (OA) movement has gained momentum, the indiscriminate growth of predatory journals has had significant negative impacts on academic communication, scholarly publishing, and effective utilization of scientific resources. This rampant growth poses a serious threat to the healthy development of the OA movement and also undermines the integrity of research and the research ecosystem. Identifying predatory journals from the massive number of OA journals would assist scholars in evading negative consequences in areas of monetary investment, reputation, academic influence, and occupational advancement. Traditional methods for identifying predatory journals have relied heavily on the knowledge of domain experts. However, a large number of predatory journals exhibit latent and covert characteristics, and the growth rate of OA journals is extremely rapid, making it difficult for experts to identify these predatory journals from the vast number of OA journals. This paper proposes an interpretable machine learning model for early warning of predatory OA journals, which identifies predatory journals through the ensemble of multiple machine learning algorithms. Specifically, the proposed methodology first constructs an OA journal early warning indicator system and integrates multiple machine learning algorithms to compute the early warning values of OA journals. Then, the SHAP interpretable framework is introduced to analyze the causal factors of the early warning risks in a novel way. To verify the accuracy of the model's causal factors, we conduct a comparative analysis of domestic and foreign medical OA journals using case studies. The empirical analysis conducted in this study demonstrates the efficacy of the ensemble algorithm in accurately identifying the risk of predatory OA journals.

Keywords Open access · Journal alerts · Causal analysis · Interpretable machine learning

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Introduction

As an emerging academic publishing model, open access (OA) has been proposed as an innovative way to facilitate academic information exchange, promote research dissemination, and advance social resource sharing. Since its proposal in 2001, OA has undergone a gradual process of transition from skepticism to recognition and widespread promotion. Concurrently, the global market for open publishing has continued to expand. As of July 31, 2022, the Directory of Open Access Journals (DOAJ) has indexed 18,049 open access journals, encompassing 7,774,206 OA articles (DOAJ, 2022). However, high article processing charges (APCs) has become increasingly prevalent among OA journals. In 2022, Clarivate (2022) Analytics reported that nearly 13,000 journals have at least one gold open access article. End-users prioritize rapid publication and free access to scholarly articles, while publishers seek to increase their impact and financial benefits through OA publishing. This strong market demand and inherent drivers of change have propelled the explosive growth of OA publishing, making it an inevitable trend for future knowledge dissemination and formal scholarly publication.

However, during the rapid expansion of OA publishing, some negative issues have surfaced due to the low barrier of entry, substantial profits, and difficulties in regulation. For example, the "Bohannon incident" revealed issues of lax review (Bohannon, 2013), a certain OA journal falsifying its editorial board (Normile, 2022), and the acceptance of papers without proper review due to the journal's focus on profit (John, 2019).These incidents have caused doubts and controversy within the academic community, industry, and readership regarding the academic quality control, fairness of peer review, and publishing standards of OA journals, resulting in a crisis of trust in the OA publishing model. To tackle these negative trends, various organizations and individuals have taken measures. For example, in 2012, Jeffrey Beall, an associate professor at the University of Colorado, USA, published a blog post listing over 1000 global predatory OA publishers and OA journal names (Butler, 2013). China values monitoring the quality of international journals, as evidenced by the efforts of numerous universities and research institutions to compile warning lists. For example, since 2020, the National Science Library of the Chinese Academy of Sciences has published the "International Journal Warning List (Trial Implementation)" for two consecutive years (NSL, 2020). Additionally, Sun Yat-sen University has created a "blacklist" of humanities and social sciences journals (The Paper, 2022). In 2022, Clarivate Analytics officially implemented the self-citation suppression criteria to address the situation where a journal's citation of its own papers is highly concentrated, artificially inflating the journal's impact factor, and issued warnings to six journals regarding this issue (Clarivate, 2022).

The allure of substantial profits has attracted a cohort of speculators to venture into the open access (OA) publishing market. Journals driven by financial gain have heedlessly expanded their article volumes and relaxed manuscript review standards. Consequently, there has been a precipitous decline in paper quality, a drain on scientific research funds in various nations, and a profound disruption to the normal progression of the OA publishing ecosystem. Considering these challenges, it is imperative to implement measures to sift through the vast array of OA journals, identifying those beset by quality issues. This will help scholars to avoid potential setbacks such as financial losses, reputational damage, loss of scholarly impact, and impeded career development.

Presently, the evaluation of OA journal quality primarily revolves around the development of an OA journal quality assessment index system (Vundavalli et al., 2016; Wei,

2019; Zhang et al., 2007), often overlooking a comprehensive exploration of other contributing factors. Regarding the modeling analysis of predatory OA journal identification, emphasis is placed on constructing theoretical frameworks and dissecting the underlying nature of the issue (Dai & Yuan, 2018; He & Xu, 2022; Yang et al., 2017), with insufficient elaboration on the interpretive aspects of model outcomes. It is worth noting that, as the accuracy of machine learning algorithms increases, their interpretability tends to decrease (Ibrahim et al., 2019). This poses limitations on the applicability of evaluation models in forecasting and forewarning instances of predatory OA journals.

In this study, building upon prior research, we address the following key facets:

- Creation of an OA journal quality index system encompassing seventeen indicators across four dimensions, coupled with a classification mechanism for OA journals based on an authoritative roster of early warning publications.
- Fusion of diverse machine learning algorithms to intelligently detect journals fraught with quality risks.
- Quantifies and assigns importance to each feature variable through the SHAP interpretation framework. This enables a detailed analysis of the root causes of early warning risks, while enhancing the interpretability of the machine learning model.

Literature review

The evaluation of journal quality and the detection of abnormal journals have consistently been focal points, exerting a substantial impact on both scholarly research and the practical publishing landscape. Particularly, OA journals have attracted much attention since their inception, signaling that knowledge dissemination and formal scholarly publishing will inevitably move towards OA in the foreseeable future. As a result, identifying anomalies in the field of OA journals has become increasingly important, which further emphasizes the need to advance the OA journals platform.

Traditional approaches to evaluating journal quality have predominantly concentrated on the analysis of individual metrics. As early as 1955, Garfield introduced the concept of the journal impact factor (JIF), which subsequently became a pivotal metric in journal assessment (Garfield, 1955). Various subsequent studies have built upon this index to enhance its utility. Later, physicist Hirsch (2005) introduced the h-index, a metric aimed at quantifying the research output of independent researchers by combining publication count and citation metrics. This concept was later extended to journal evaluation (Bornmann & Daniel, 2005). Another metric, the Source Normalized Impact per Paper (SNIP), was devised to mitigate citation behavior discrepancies across distinct fields by normalizing the average citations per article. SNIP, proposed by Professor Moed, emerged as a sophisticated indicator of journal citation impact (Moed, 2011). Furthermore, the SCImago Journal Rank (SJR), introduced in 2008, incorporates both citation volume and the quality of citations. It employs an algorithm akin to PageRank and offers an alternative to the impact factor (Butler, 2008; Falagas et al., 2008). In an endeavor to provide a rival to the JIF, Elsevier introduced the CiteScore (CS) metric in 2016. These metrics collectively aim to establish a unifying principle for academic assessment. It's worth noting that the application of different metrics can yield disparate rankings, underscoring the intricate nature of journal evaluation. In the field of predatory journal identification, Dadkhah and Bianciardi (2016)

proposes a brand new metric, Predatory Rate, which helps to make judgements about predatory journals and has some practical significance.

The study also explored the issue of combining individual indicators, including the use of multiple indicators to assess journals. The advantage of combining different indicators is that it reduces one-sided biases and errors that may exist in a single indicator, thus contributing to a more comprehensive assessment of journals. For instance, Wu et al. (2015) formulated an encompassing index framework for the citation evaluation of scientific and technical journals, incorporating five pivotal indicators: impact factor, h index, SJR, Eigenfactor Score, and SNIP. This framework serves as a point of reference for appraising the current landscape of scientific and technical journals. Dai and Yuan (2018) established a reputation risk assessment mechanism tailored to OA journals, which revolves around influence and is rooted in the reputation incentive mechanism proposed by Kelvin. This framework incorporates multiple indicators, including impact factor, cited half-life, and h5-index, to quantify reputation risk. Zhao and Wang (2019) devised a multi-indicator fused evaluation index system spanning academic and social dimensions. This system facilitated a comparative analysis of the impact of OA journals in China and the United States. Employing statistical methodologies, the authors examined and interpreted the evaluation index system. Furthermore, statistical analyses were conducted by Su et al. (2015) on the data derived from the Journal Citation Reports (JCR) and Science Citation Index (SCI) papers. This study employed ten indicators, encompassing aspects such as language, to evaluate journals focusing on smart city construction and building technology. The findings underscored the significance of these indicators in assessing journal quality and impact.

The construction of the multi-indicator evaluation system alleviates the inaccuracy of a single indicator to a certain extent. Nonetheless, the manual selection of multidimensional indicators is manually screened with a large amount of workload and subjectivity. This challenge can be effectively addressed through the application of machine learning methodologies. In recent years, a growing body of scholarly work has endeavored to leverage intelligent algorithms, including machine learning techniques, for the discernment of predatory journals (He & Xu, 2022; Yang et al., 2017). Ma et al. (2022) proposed a time-series analysis framework for the comprehensive and dynamic evaluation of scientific and technological journals. Within this context, they established a multivariate short-time multi-series time series dataset, encompassing 18 journal evaluation indicators. The evaluation of journals can be framed as a classification problem, thereby facilitating the application of machine learning algorithms. Similarly, Yu et al. (2022) concentrated on the bibliometric indicators of economics journals in the JCR 2019 dataset. Employing K-means clustering, they derived classifications for the evaluation outcomes. Subsequently, pivotal journal evaluation indicators were computed through principal component analysis and partial least squares discriminant analysis.

Recently, several new approaches have been proposed to measure the contribution of features to machine learning models. With the introduction of the concept of interpretability, how black-box models such as machine learning make decisions can be assessed and validated at a finer granularity. While ensuring a high level of learning performance, it enables the construction of models that are more in line with people's perceptions and ways of thinking, guiding them to make decisions. Interpretable machine learning techniques provide viable solutions for revealing the mechanisms by which features affect model performance. These techniques have recently been adopted by several studies in the field of scientometrics. For example, some studies have combined features such as paper text and metadata to predict scholarly citations using algorithms such as random forests and neural

networks, and have analysed models using a variety of interpretable methods, such as feature significance, LIME, and SHAP, which in turn reveal the mechanisms that influence the citation of a paper (Beranová et al., 2022; Ding & Ruan, 2022). There are also studies that apply interpretable machine learning to the identification of high-quality patents and use the SHAP interpretable framework to explore the important factors affecting patent quality (Fu et al., 2022; Sun et al., 2022). In addition, Tian et al. (2023) developed an event-based group evolution prediction task in order to track the evolution of scientific communities and delve into the mechanisms behind them, and combined algorithms such as XGBoost with SHAP to provide a solution for revealing the underlying mechanisms of the evolution of scientific communities.

However, no interpretable analyses of risk warning links in OA journals have yet been conducted. Our research endeavours to incorporate a series of sophisticated machine learning algorithms to enable intelligent detection of OA journals with quality risks. In order to improve the credibility of the results, we attempt to use the SHAP explanatory framework to quantify and attribute the significance of each feature variable, providing a robust method to examine the complex relationships between input variables and output predictions, thereby enhancing the interpretability of the overall model.

Methodology

Research framework

To enhance the academic influence of journals and mitigate the risk of OA journals entering the early warning list, this study proposes an interpretable model that integrates XGBoost and SHAP for OA journal risk early warning. The ensemble learning approach has demonstrated significant effectiveness; however, it has yet to fully address the issue of interpretability. Historically, when employing the integrated model for classification tasks, the model's output manifested as a probability value. In this context, the Shapley value emerges as a remedy rooted in cooperative game theory, advanced by the eminent scholar Shapley (1953). By adapting this concept to the realm of machine learning (Lundberg & Lee, 2017; Ribeiro et al., 2016), the Shapley value of each feature can be meticulously computed. This calculation serves to gauge the impact of individual feature values on the ultimate output, thereby facilitating the pursuit of interpretive objectives.

The model employs machine learning techniques to construct an OA journal risk early warning model, integrates the SHAP explanatory framework to quantify and attribute the importance of each characteristic variable, and provides an explanation of the crucial factors affecting the formation of OA journal risk and their mode of operation. The model's efficacy is verified through a quantitative comparative analysis of medical OA journals and engineering OA journals based on risk-causing indicators. Figure 1 illustrates the framework of the model study.

(1) Module 1: Data acquisition and pre-processing. Transforming and normalizing the data collected from various websites, as well as addressing missing values through appropriate handling methods.
(2) Module 2: Indicator system construction. The Pearson correlation coefficient was utilized to calculate the correlation between variables, while the eXtreme Gradient Boosting (XGBoost) algorithm was employed to calculate the feature importance score. By

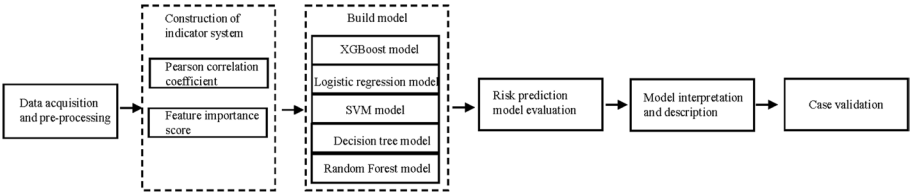


Fig. 1 Predatory OA Journal Risk Warning Model Based on Interpretable Machine Learning

integrating the results of the two methods, feature variables were extracted, and any irrelevant variables were eliminated from the journal data to form the dataset.

(3) Module 3: Model construction and evaluation. The model was compared with four mainstream machine learning algorithms, namely Logistic Regression (LR), Support Vector Machines (SVM), Decision Tree, and Random Forests. The evaluation of the model was based on four performance metrics, namely Accuracy, Recall, F1score, and Area Under Curve (AUC), in order to identify the optimal model.

(4) Module 4: Model interpretation and explanation. The SHAP explanatory framework was introduced to visually and conceptually explicate the optimal integrated model, thereby facilitating the identification of the critical factors that influence the formation of risk in OA journals, as well as their operational mechanisms.

(5) Module 5: Validation of applicability. The applicability of this risk control model was validated through a quantitative comparative analysis of medical OA journals based on risk-causing indicators.

Data acquisition and pre-processing

The data sources for this study included database platforms such as Wos, Scopus, MedSci, CWTS, and Scimago. The study focused on medical OA and engineering OA journals listed in the International Journal Alert List (Trial) published by the Literature and Information Center of the Chinese Academy of Sciences during the years 2020–2023. After collecting journal titles, we searched for metrics data across platforms based on the ISSN numbers of the corresponding journals in order to form the final dataset.

Indicator system construction

Yu et al. (2023) believes that both journal timeliness and journal influence are important bases for comprehensive evaluation of journals, and uses this to construct a model of four-quadrant partitioning. As an emerging publishing mode, many systems of OA journals are not yet sound, and Dai and Yuan (2018) conduct an assessment study on the academic reputation risk of OA journals, and proposes an analysis model containing ten indicators in four dimensions, such as the number of articles carried and the self-citation rate et al.. For evaluating journal quality, there are widely used indicator systems, such as the JCR indicator system introduced by Clarivate Analytics and the indicator system based on the Scoups database introduced by Elsevier. From the predatory OA journals to achieve the purpose, the first thing to make the journal can enter the SCI or Scoups list, and usually use covert means to quickly improve the value of the impact indicators of these journals, disguised as a high-quality journal, to attract more authors to submit manuscripts. Therefore,

the identification of predatory OA journals still has to examine these conventional journal indicators, which are summarized in this paper as the first dimension: the journal impact dimension. Second, predatory OA journals can enhance the value of journal quality indicators in the short term by means of manipulation, but, in the long term, truly quality papers will show sustained citations over a long period of time; therefore, this paper classifies indicators that measure journal timeliness, such as cited half-life, as the second dimension: journal impact timeliness dimension; third, predatory OA journals originate from the open access movement, which, in order to be faster and more enrichment, they will inevitably speed up the publication cycle and increase the manuscript size, therefore, this paper lists journal external indicators such as review cycle and contribution hit rate as the third dimension: journal characteristics dimension; fourth, usually predatory OA journals focus on a certain range in order to reduce the cost of publicity, which results in the tendency of a more concentrated geographic location of the paper's authors, for example, Bohannon found that predatory journals are concentrated in developing countries, therefore, the proportion of single-country issuances is also included in the scope of examination as the fourth dimension: the international cooperation dimension. The meanings of these four dimensions and their indicator variables are detailed in Table 1.

To enhance the validity of the feature model, it is essential to conduct correlation analysis on the feature variables and remove irrelevant variables with low correlation to the public value consensus. In this study, the Pearson correlation coefficient and XGBoost-based feature importance score were utilized to objectively demonstrate the structural level of the evaluation model and extract secondary indicators. The Pearson correlation coefficient was calculated as follows:

$$\rho(X, Y) = \frac{Cov(X, Y)}{\sigma(X)\sigma(Y)} \tag{1}$$

whereas $\sigma(X)$ and $\sigma(Y)$ represent the standard deviations of X and Y, respectively, and $Cov(X, Y)$ represents the covariance of X and Y, the Pearson correlation coefficient is indicated by r and assumes values ranging between -1 and 1, with 0 signifying no correlation. The absolute value of the correlation coefficient is indicative of the strength of correlation, with values closer to 1 denoting higher correlation. In the XGBoost algorithm, the feature importance metrics are computed as follows: AverageGain signifies the average gain of the feature in all trees; AverageCover represents the average number of samples covered by the feature in all trees; and Fscore denotes the percentage of times the feature is utilized in the split nodes, which is determined using formulas 2 and 3. This study employs Fscore as the criterion for calculating the importance score.

$$Averrage\ Gain = \frac{\sum Gain}{F\ score} \tag{2}$$

$$Average\ Cover = \frac{\sum Cover}{F\ score} \tag{3}$$

Model construction and evaluation

By utilizing the relevant feature variables identified earlier as characteristic variables and journal risk as the predictor variable, an XGBoost-based OA journal risk warning model was

Table 1 Dimensions of OA journal risk prediction models and corresponding indicators

Dimensions	Indicator variables	Variant interpretation	Reference research	Note
Journal impact strength	Impact Factor	The total number of citations to papers published in a journal in the previous two years in that reporting year (JCRyear) divided by the total number of papers published in that journal in those two years	Zarifmahmoudi et al. (2015), Sánchez et al. (2019)	In general, the larger the indicator for this dimension, the more influential the journal is in the field as a whole
	5 Year Impact Factor	The ratio of the total number of citations of papers published in a journal in the previous five years to the total number of papers published in the journal in the previous five years in the year of enumeration	Paji (2015)	
	JIF without self cites	Impact factor of a journal after removal of self-citation	Valderrama et al. (2020)	
	Eigenfactor score	is the ratio of the number of citations to the total number of papers	Cantín et al. (2015)	
	Immediacy Index	An indicator of the immediate response rate of a journal, mainly in terms of how many papers published in the journal was cited in the year	Sánchez et al. (2019)	
	Article influence score	Eigenfactor divided by the normalized ratio of papers published in the journal	Paji (2015)	
	SJR	Considers both the number of citations and the quality of citations of a journal	Ahmad and Waris (2017)	
	SNIP	The ratio of the average crude impact per page (RIP, i.e., 3-year impact factor) of journal publications to the relative citation potential (RDCP) of databases in the journal's subject area	John and Yang (2017)	
	Normalized Eigenfactor	Standardized results for feature factors	Jaafar et al. (2021)	

Table 1 (continued)

Dimensions	Indicator variables	Variant interpretation	Reference research	Note
Journal Impact Timeliness	Cited Half-Life	The time span over which the more recent half of all citations to a journal paper in a given year were published	Halim and Khan (2019)	The larger the indicator for this dimension, the more influential the journal is in terms of citations
	Citing Half-Life	How long has the newer half of all references in a journal been published	Paji (2015)	
	CiteScore	The total number of citations for papers published in a journal in the previous three years in the year of enumeration divided by the total number of papers published in the journal in the previous three years	Halim and Khan (2019)	
Journal Features	Review cycle	The time period between submission and review and when the journal notifies you that the article has been accepted and sends you a notification of acceptance	Hu et al. (2010)	All of these indicators need to be within an appropriate range, and both too high and too low need to be taken seriously
	Contribution Hit Rate	Probability of acceptance of an author's submission to a journal	Mo and Ma (2012)	
	Volume of articles	The number of papers in related disciplines published in a given journal in a given period	Halim and Khan (2019)	
	Self-citation rate	The proportion of self-citations to the total citations of a journal in a given period	Liu et al. (2011)	
International cooperation	Percentage of single country issuances	The proportion of articles published in a journal from a particular country	Zong (2022)	A very large value indicates that there may be an anomaly in the journal

developed. Four widely-used mainstream machine learning models, namely logistic regression (LR), support vector machines (SVM), Decision Tree, and Random Forests, were chosen as comparison models. The selection was based on the following considerations:

The XGBoost algorithm is a decision tree-based algorithm that employs gradient boosting and falls under the category of integrated approaches. However, it differs from other algorithms in terms of the cut-off strategy and its support for categorical features. In this type of algorithm, decision trees are linked together, and the current prediction is based on the prediction error of the previous round. The prediction error of each round is utilized to iteratively construct the model and enhance the prediction accuracy (Luan et al., 2020). It holds a distinct advantage over the conventional Gradient Boosting Decision Tree (GBDT) due to its incorporation of second-order Taylor expansion. This strategic inclusion effectively minimizes model parameters, resulting in heightened evaluation precision. Bolstered by its robust fitting capability, this method adeptly derives intricate models from data. Moreover, its computational process has garnered notable acclaim, finding widespread application in data science and various machine learning competitions. Remarkably commendable outcomes have been achieved through this approach.

Logistic regression (LR) is a generalized regression model that can be applied to multi-classification and dichotomous problems, with the benefit of high interpretability and low complexity. Support vector machines (SVM) is a generalized linear classifier for supervised learning, utilized in binary data classification, with exceptional non-linear representation and multi-layer feature learning capabilities. The Decision Tree algorithm is a method for approximating discrete-valued functions, using a "top-down, divide and conquer" approach to partition the search space into several disjoint subsets for classification and prediction. While fast, it is prone to overfitting. Random Forest, on the other hand, is a representative algorithm of Bagging that uses decision trees as a base learner to construct an integrated algorithm. It excels in dealing with high-dimensional, feature-missing, and unbalanced data, and can prevent overfitting when dealing with noisy data.

The five classification models were trained and predicted by dividing the dataset into a 70% training set and a 30% test set. To determine the overall best model, accuracy, recall, F1-score, and AUC were utilized as performance indicators. The formulas for these indicators are provided as follows:

$$Accuracy = \frac{T_P + T_N}{T_P + F_P + T_N + T_N} \tag{4}$$

$$Recall = \frac{T_P}{T_P + F_N} \tag{5}$$

$$F_1\text{–}Score = \frac{2T_P}{2T_P + F_N + F_P} \tag{6}$$

where T_P represents the number of correctly predicted standard samples, F_p represents the number of mispredicted normal samples, T_N represents the number of correctly predicted abnormal samples, and F_N represents the number of mispredicted abnormal samples.

Model interpretation and explanation

The model is designed to be post-hoc interpretable and leverages the SHAP interpretable technique, which was developed by Lundberg and Lee (2017). The technique provides a solution to the machine learning black box problem by offering both local and global interpretability. It calculates the average marginal contribution of a sequence of features and considers the impact of individual features, as well as two-way feature crossovers, while addressing the problem of multicollinearity. This framework is a valuable tool for understanding the impact of various factors on the model's predictions. The formula for calculating the SHAP value is shown as follows:

$$\phi_i = \sum_{S \subseteq N\{i\}} \frac{|S|!(M - |S| - 1)!}{M!} \cdot [v(S \cup \{i\}) - v(S)] \tag{7}$$

In this context, we define ϕ_i as the contribution of the first i indicator, S as the subset of indicators, $N\{i\}$ as the set of all indicators, and M as the total number of input indicators. Additionally, $v(S \cup \{i\})$ represents the predicted value of the model when the sample has only the middle indicator value $S \cup \{i\}$, while $v(S)$ represents the predicted value of the model when the sample has only the middle indicator value. Based on the additive characteristic property method g, define the linear function, i.e.

$$g(x) = \phi_0 + \sum_{i=1}^{M} \phi_i x_i \tag{8}$$

Formula 8 provides the post-hoc interpretation model prediction,where $g(x)$ represents the predicted value for the sample i, ϕ_0 is the average value of the model's predicted values, and x_i denotes the i sample feature. For a prediction problem involving three input variables, a schematic model of formula 8 is illustrated in Fig. 2, with red indicating a positive SHAP value and blue indicates a negative SHAP value. Figure 2 depicts how the SHAP value denotes the contribution of each transporter variable towards predicting a sample (Feng & Wu, 2022). Here, ϕ_1 denotes the positive impact of variable one in predicting changes, which increases the current prediction. Likewise, the addition of variable two, represented by ϕ_2, leads to an increase in the current prediction. However, introducing variable three results in a decrease of the current prediction to ϕ_3.

The SHAP explanatory framework was employed to visualize and interpret the comprehensive optimal model, with the aim of analyzing the critical factors that influence the formation of risk in OA journals and demonstrating how each feature affects the prediction results. Three SHAP visualization diagrams were utilized for this purpose: (1) The

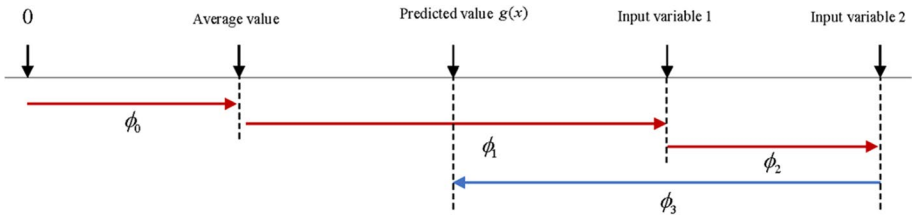


Fig. 2 SHAP method with three input variables

SHAP feature importance bar chart was used to reflect the degree of influence of features on the model and to identify the feature variables that have a more significant influence on the formation of OA journal risk; (2) A SHAP summary chart was drawn to depict the overall positive and negative relationships between feature variables and OA journal risk formation, clarifying the direction of influence of feature variables; (3) The SHAP partial dependency chart was employed to analyze the marginal effects of each feature and how the specific values of each feature variable affect the formation of OA journal risk, thus providing a clear understanding of the influence of feature variables and their strength.

Results

Data acquisition and pre-processing

OA journals can be categorized into Full OA journals and Hybrid OA journals. In Full OA journals, authors or their institutions pay the APC, and readers can access the articles for free after publication. On the other hand, Hybrid OA journals combine traditional and OA modes of operation.

To account for differences in disciplines and the number of journals in the early warning list within the same discipline, this study selected medical OA and Hybrid OA journals from the International Journal Early Warning List (Trial) published by the Documentation and Information Center of the Chinese Academy of Sciences between 2020 and 2023. An equal number of medical OA and Hybrid OA journals were randomly selected from each of the four divisions of SJR, and these served as the normal sample of standard journals.

The annual indicators of Impact Factor, Volume of Articles, Immediacy Index, Cited Half-Life, and CiteScore were obtained from the WoS and Scopus databases. The review cycle was obtained by calculating the average review time of 10 papers under each journal. Contribution Hit Rate was obtained from the MedSci database. The annual indicator for SNIP was obtained from the CWTS website at www.journalindicators.com/indicators, and the annual indicator for SJR was obtained from the Scimago website at www.scimagojr.com/journalsearch.php. The annual indicator for the percentage of single country issuances was obtained by querying the WoS database and calculating the number of articles published in the journal from a particular country in that year divided by the annual number of articles carried.

The ISSN numbers of the journals were used to search for the above indicators and data for the period 2013–2022. Any missing data were excluded, and this resulted in a panel of 434 data of medical journals and 194 data of engineering journals with a positive-to-negative sample ratio of 1:1. Considering that differences in disciplines may have an impact on the calculation results, we do separate experiments with data from medical journals and engineering journals, thus helping us to observe the differential characteristics of early warning journals in different disciplines. The basic information of the numerical features is depicter in Table 2.

Indicator system construction

The process of constructing the indicator system is designed to eliminate multicollinearity between indicators and remove indicators with high correlation. Common methods to eliminate multicollinearity are principal component analysis (PCA) and regularisation

Table 2 Summary statistics of numeric features

Feature	Max (m)	Max(e)	Min (m)	Min (e)	Mean (m)	Mean (e)	SD (m)	SD (e)
IF	39.890	15.204	0.509	0.380	4.238	3.531	4.278	2.352
5IF	38.658	16.936	0.434	0.611	4.271	3.673	4.181	2.413
JWSC	38.590	14.981	0.182	0.501	3.943	2.920	4.120	2.121
ES	0.193	0.124	0	0	0.021	0.022	0.032	0.026
II	60.269	3.008	0.056	0.077	1.295	0.821	3.441	0.626
AIS	15.274	5.126	0.052	0.143	1.301	0.739	1.817	0.696
SJR	9.631	4.703	0.170	0.240	1.489	1.073	1.426	0.701
SNIP	5.520	5.670	0.170	0.322	1.298	1.507	0.814	0.793
NE	31.529	26.785	0.010	0.062	2.852	3.403	4.391	4.746
CHED	13.900	11.200	1.400	1.500	5.889	5.044	2.886	2.559
CH	11.500	12.300	3.500	4.300	7.037	7.094	1.260	1.608
CS	25.300	21.500	0.800	0.700	5.786	5.531	4.441	3.713
RC	8.800	12.800	1.000	1	2.851	4.612	1.529	3.268
CHR	0.950	0.950	0.083	0.250	0.473	0.649	0.160	0.172
VOA	9428	10,907	7	35	480.051	1098.489	809.559	1584.481
SCR	0.750	0.500	0	0.010	0.092	0.188	0.110	0.098
POSCI	0.909	0.830	0	0	0.256	0.296	0.278	0.178

methods. Dong and Bollen (2015) designed a consumer confidence calculation model based on large-scale online attention data, using PCA to eliminate the correlation between indicators. The regularisation methods contain L1 (lasso regression) and L2 (ridge regression). Huang et al. (2016) developed radiomics nomogram for preoperative prediction of lymph node (LN) metastasis. Data dimensionality reduction, feature selection and radiomics feature construction using Lasso regression model. The Pearson correlation coefficient was utilized to calculate the correlation among the indicators, and the heatmap results are presented in Fig. 3. The importance score of each indicator was calculated using XGBoost, and the importance scores of the selected indicators are displayed in Table 3. The Pearson correlation coefficients and importance scores of the journal indicators were combined with the specific significance of the indicators. Indicators with a correlation of ≥ 0.8 and low importance scores were excluded, resulting in 12 relevant variables: Impact Factor, Immediacy Index, Volume of articles, Cited Half-Life, Citing Half-Life, Eigenfactor Score, Self-citation rate, SJR, SNIP, Review cycle, Contribution Hit Rate, POSCI. The engineering data was also filtered to obtain 11 relevant variables: Immediacy Index, Volume of articles, Cited Half-Life, Citing Half-Life, Eigenfactor Score, Self-citation rate, SJR, SNIP, Review cycle, Contribution Hit Rate, Percentage of single country issuances.

Model construction and evaluation

The study utilized 11 relevant variables categorized under four dimensions, namely journal influence, timeliness, characteristics, and international cooperation, as feature variables, with OA journal risk as the predictor variable. Five classification models, namely logistic regression, support vector machine, decision tree, random forest, and XGBoost models,

Fig. 3 The correlation matrix of features. **a** The correlation matrix of medical journal data. **b** The correlation matrix of engineering journal data. *Note* The '×' in the middle of the block indicates that the correlation between the two features is not significant, (i.e., $p > 0.05$); and if there is no '×' in the block, this means that the two features are significantly correlated

were developed using a dataset of 434 medical journals data and 194 engineering journals data, where 70% of the data served as the training set, and 30% as the test set.

Accuracy, precision, recall, F1-score, and AUC serve as metrics to assess the strengths and limitations of the model. Accuracy denotes the ratio of correctly classified samples to the total number of samples, whereas recall measures the percentage of positive samples that are correctly identified as such. The combined metrics of F1-score and AUC are also used. The test outcomes following a five-fold cross-validation and grid search are tabulated in Table 4, where the best and second-best values are displayed in bold respectively, and the best value is underlined. As shown in Table 4, all models obtained excellent scores exceeding 0.80 for Accuracy, Recall, and F1-Score, signifying that the metrics are appropriately constructed and can better explain the reasons for early warning risks in OA journals. The ROC plots for each model are shown in Fig. 4. We also tried mixing the two data as a new dataset for the above mentioned operations and the results are also presented in Table 4.

Model interpretation and explanation

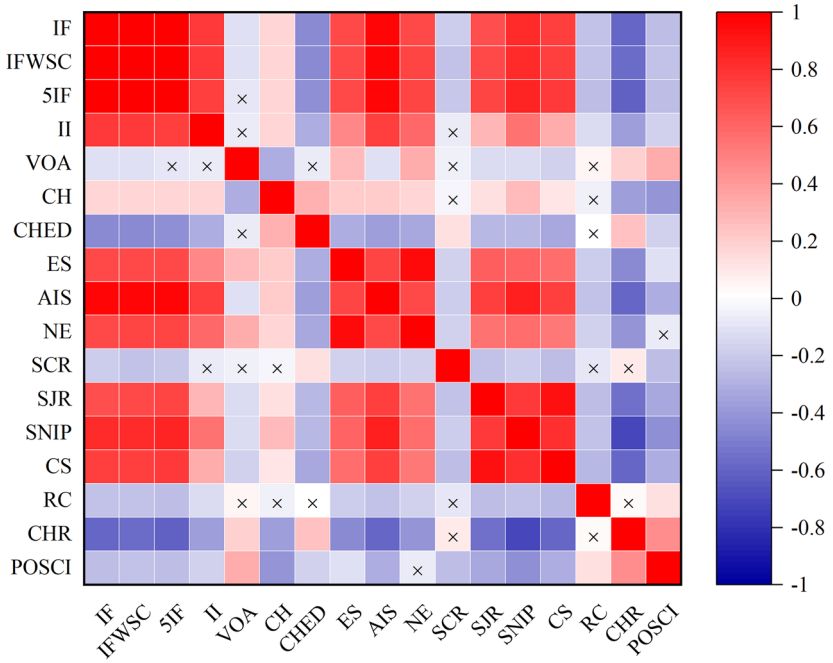
Importance analysis of characteristic variables

SHAP feature importance quantifies the contribution of each feature towards enhancing the predictive ability of the model and is measured by the absolute value of the degree of influence of each feature on the target variable (Abp et al., 2020). It offers a graphical depiction of the level of influence that each feature has on the model. The importance of a feature variable is directly proportional to its impact on the development of risk in OA journals. Consequently, the higher the importance of a feature variable, the greater its influence on the model.

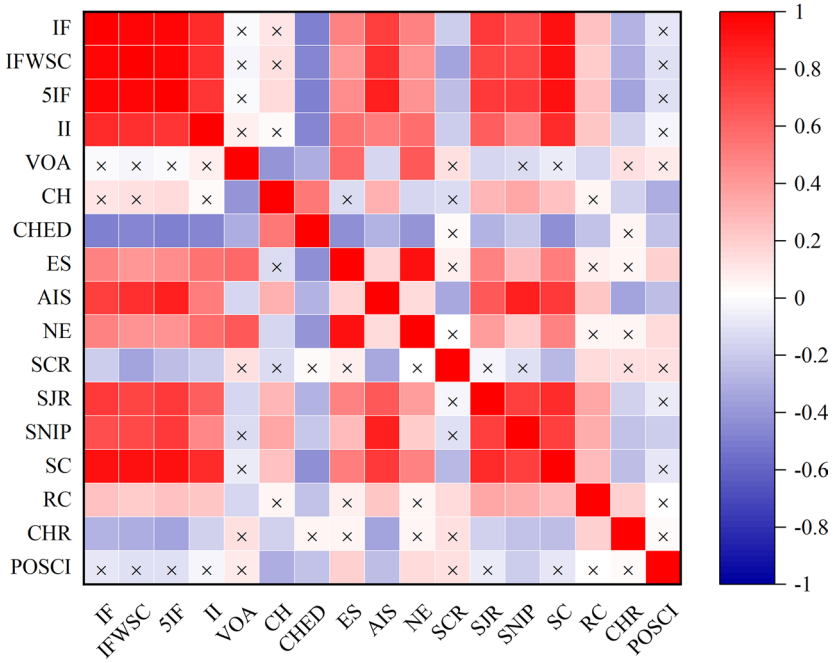
Figure 5a illustrates the descending order of importance for the characteristic variables, which is as follows: percentage of single country issuances, Cited Half-Life, Review cycle, SNIP, Contribution Hit Rate, and SJR. Notably, the percentage of single country issuances exerts a substantial impact on model prediction and is a crucial indicator for constructing risk prediction models for OA journals.

As shown in Fig. 5b, the characteristic variables of engineering journals are, in descending order of importance, SJR, SNIP, Cited Half-Life, Self-Citation Rate, Citing Half-Life, Immediacy Index, and Review Cycle.

The most important difference between the two types of journal indicators is the "SJR", which is an indicator that takes into account both the number of citations and the quality of citations to the journal, but because journals have different attractions for researchers in different fields and the quality of journals in different fields is evaluated according to different criteria, this factor should be taken into account to filter this indicator in the construction of the final model. This factor should be taken into account when constructing the final model to filter this indicator. Other than that, the importance of the other indicators does not vary much. The major difference between these two types of journal indicators are



(a) The correlation matrix of medical journal data



(b) The correlation matrix of engineering journal data

Table 3 Importance scores of selected indicator

Medical Journal		Engineering Journal	
Indicator	Importance score	Indicator	Importance score
Percentage of single country issuances	0.3358	SJR	0.4540
Self-citation rate	0.1613	SNIP	0.0842
Contribution Hit Rate	0.0934	Citing Half-Life	0.0835
Eigenfactor Score	0.0653	Self-citation rate	0.0733
Impact Factor	0.0648	Immediate Index	0.0685
SNIP	0.0445	Review cycle	0.0601
Review cycle	0.0420	Eigenfactor Score	0.0391
5 Year Impact Factor	0.0333	Cited Half-Life	0.0376

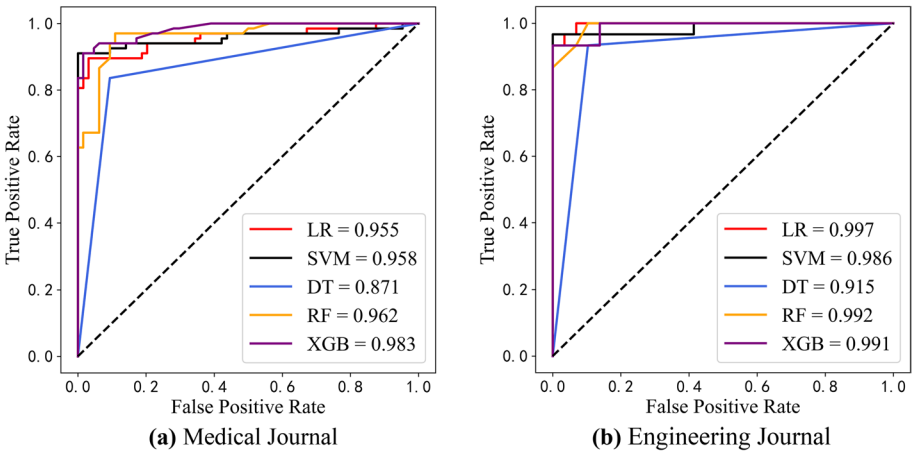


Fig. 4 ROC of the five machine learning models using datasets Medical Journal, Engineering Journal. **a** Medical Journal, **b** Engineering Journal

SJR and POSCI. The SJR takes into account both the number of citations and the quality of citations for a journal. It has different appeal to researchers in different fields, while the quality of journals in different fields is evaluated by different criteria, and POSCI may be due to disciplines leading to a different share of publications in a single country. Therefore, these factors should be taken into account when constructing the final model to filter the indicator.

Analysis of the direction of influence of the characteristic variables

The SHAP Summary Plot, as depicted in Fig. 6, integrates feature importance with feature effects to demonstrate the positive and negative associations between feature values and the formation of risk in OA journals (Abp et al., 2020). Each point on the plot represents a sample, with the red and blue colors representing high and low feature values, respectively. The SHAP value on the horizontal axis is used to gauge the extent to which each indicator

Table 4 Prediction performance of 5 machine learning models

	Models	Accuracy	Recall	F1-Score	AUC
Medical	Logistic Regression	0.924	0.896	0.923	0.955
	SVM	**0.947**	**0.910**	**0.946**	0.958
	Decision Tree	0.870	0.835	0.868	0.871
	RandomForest	0.885	0.835	0.882	**0.962**
	XGBoost	**0.939**	**0.925**	**0.939**	**0.983**
Engineering	Logistic Regression	**0.949**	**0.933**	**0.949**	**0.997**
	SVM	**0.983**	**0.967**	**0.983**	0.986
	Decision Tree	0.915	**0.933**	0.918	0.915
	RandomForest	0.932	**0.933**	0.933	**0.992**
	XGBoost	**0.949**	**0.933**	**0.949**	0.991
Mixed	Logistic Regression	0.941	0.908	0.941	0.975
	SVM	0.936	0.897	0.936	0.964
	Decision Tree	0.862	0.735	0.847	0.881
	RandomForest	**0.968**	**0.949**	**0.969**	**0.999**
	XGBoost	**0.979**	**0.959**	**0.979**	**0.990**

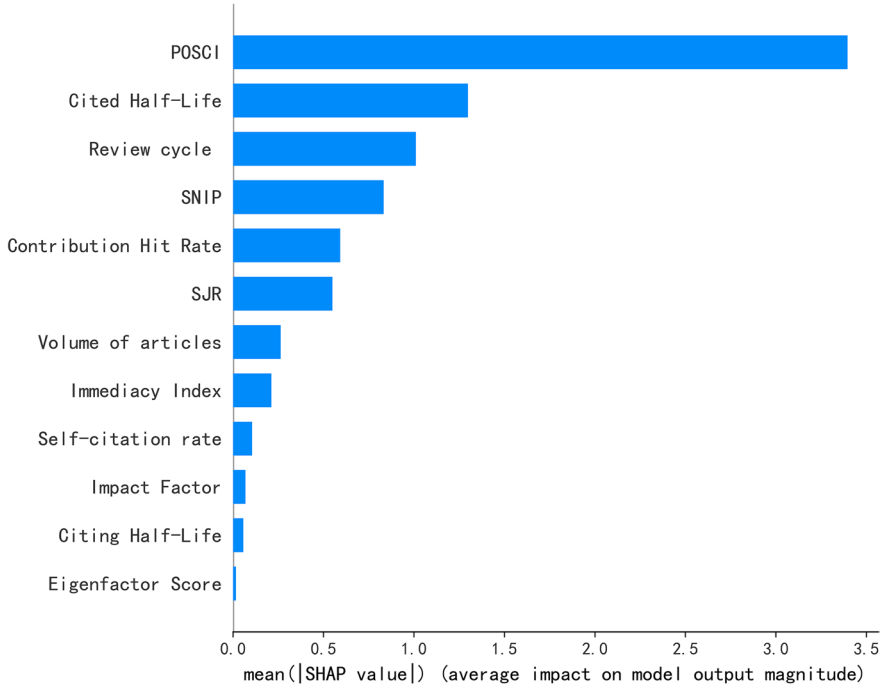
contributes to and influences the model's predicted value. When SHAP > 0, it indicates that the indicator raises the risk of the journal entering the warning, while SHAP < 0, it indicates that the indicator makes it less risky.

As shown in Fig. 6a, a decrease in the SHAP value corresponds to a decrease in the percentage of single country issuances. This implies that a journal that publishes a higher proportion of papers from countries other than its own is more likely to be categorized as an early warning journal. Moreover, higher indicators, such as SNIP, Cited Half-Life, and CiteScore, decrease the risk of a journal triggering an early warning. Conversely, higher indicators such as Self-citation rate and Contribution Hit Rate increase the likelihood of a journal being classified as an early warning journal.

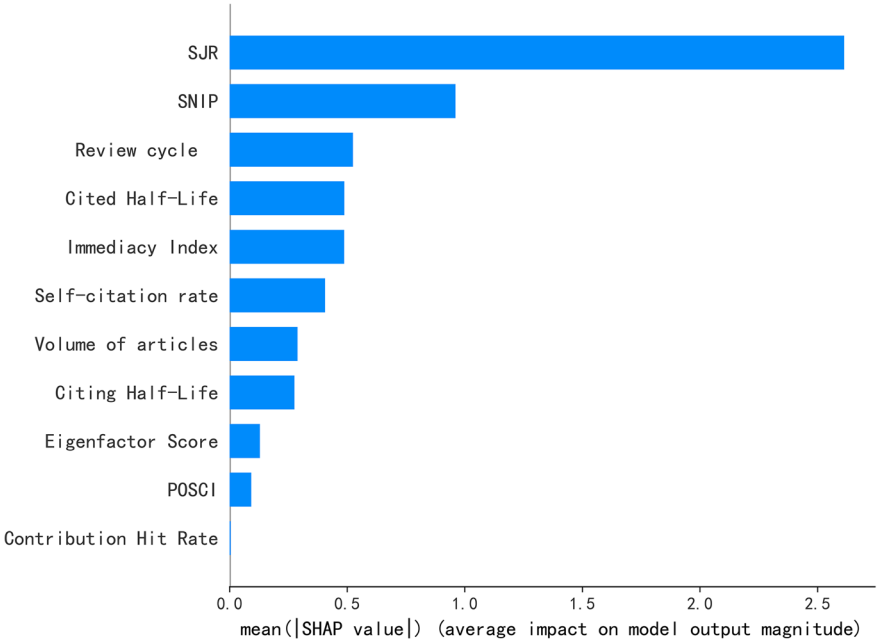
Figure 6b shows that the SHAP values vary with the magnitude of the SJR values, and as the SJR gets lower, the risk of a journal becoming an early warning journal gets higher. There is no difference in the trend of the extent to which the metrics contribute to the predicted values of the model as shown in Fig. 6a and b.

Analysis of the impact of specific values of the characteristic variables

The risk of OA journal alerts is influenced by various factors, and analyzing a single indicator alone cannot provide a thorough understanding of the causal outcomes of journal alert risk. Therefore, it is necessary to further explore the impact of multiple factors on journal alert. From the calculations, it is clear that disciplinary variability does not have a significant impact on the results; therefore, the subsequent focus of this paper on exploring medical journals will result in results that are largely applicable to engineering journals. Given the numerous combinations of interactive features and the significant influence of the percentage of single country publications on model prediction, this study focuses on the percentage of single country publications, as well as citation half-life, SNIP, review cycle, and self-citation rate as examples. To reveal the interaction of these indicators, SHAP dependency plots provide the marginal effect of the feature variable on the risk formation

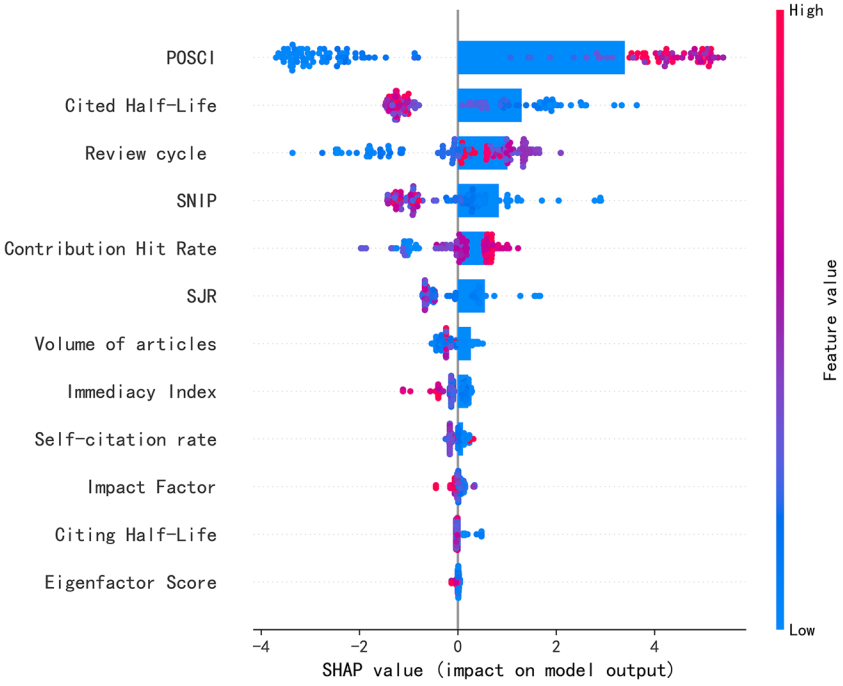


(a) Medical Journal

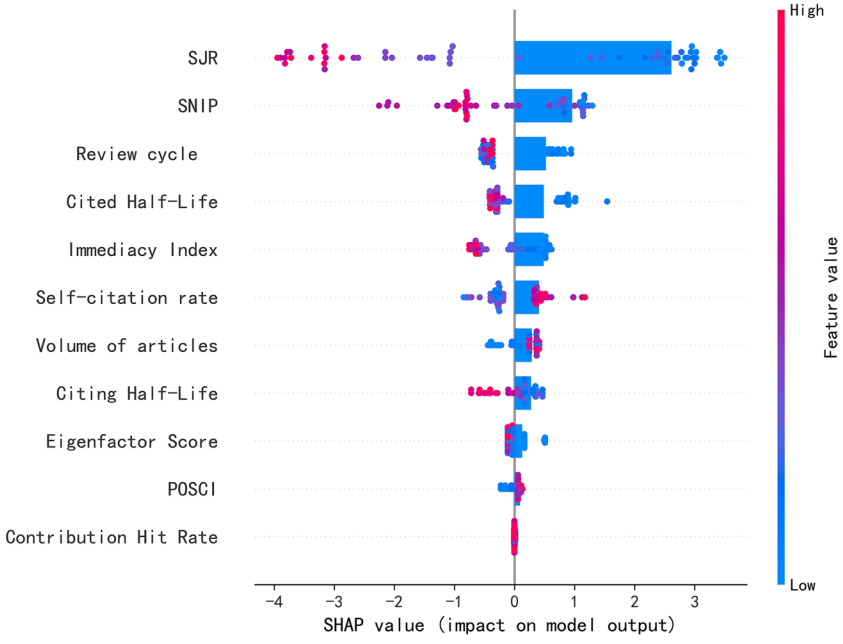


(b) Engineering Journal

Fig. 5 Importance analysis of characteristic variables. **a** Medical Journal. **b** Engineering Journal



(a) Medical Journal



(b) Engineering Journal

Fig. 6 SHAP summary plot. **a** Medical Journal. **b** Engineering Journal

of OA journals. The dispersion in the vertical direction of the percentage of single country issuances in the horizontal coordinate indicates the interaction with another feature. The vertical axis (left) indicates the percentage of single country issuances corresponding to the SHAP value for one sample. Conversely, the more extensive the value of the other feature in the vertical axis (right), the redder it appears.

Figure 7a–d depict the relationship between Cited Half-Life, SNIP, Review Cycle, Self-citation rate, and their combination with the percentage of single country issuances indicator and the prediction of early warning for OA journals. Figure 7a and c demonstrate that many journals with a high percentage of single-country publications also exhibit a favorable cited half-life and a lengthy review cycle. Figure 7b and d reveal that OA journals with a higher proportion of single-country publications are likely to have a lower SNIP and self-citation rate, and only a few journals exhibit higher SNIP and self-citation rate.

Validation of the applicability of the risk control model: a case study based on medical OA journals

In order to verify the accuracy of the models constructed in the previous sections, we collected indicator data from journals ABC, which were included in the early warning list in 2023, and normal journals DEF, which were used as inputs to the model, compared the model predictions with the actual situation, and analysed the final results using an interpretable model. Since the indicator predictions have a lag, we selected the data input to the

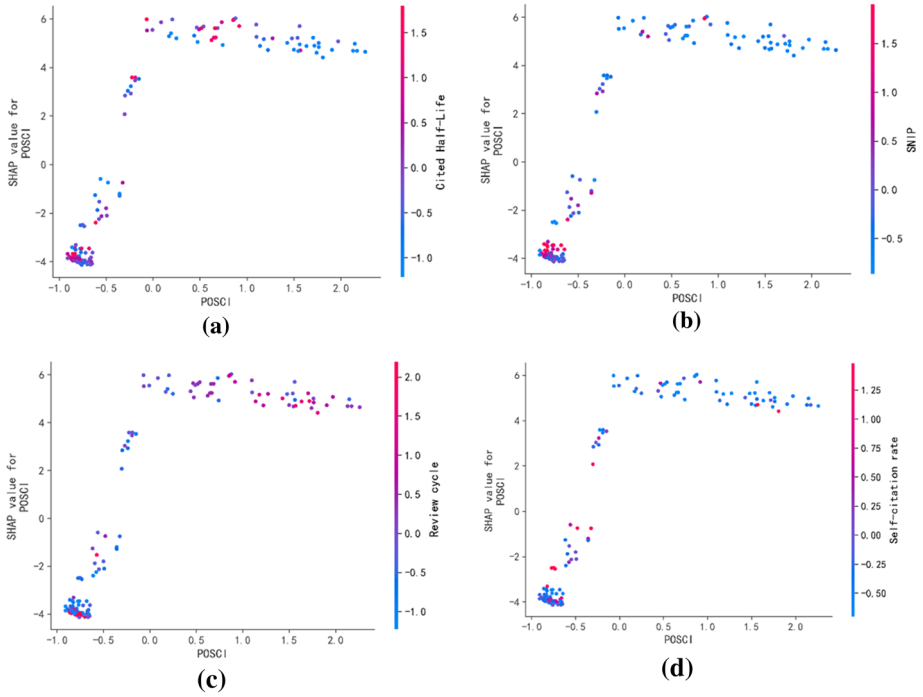


Fig. 7 Feature dependence analysis

model, both from 2022 and 2021. The journals AB and DE are journals in the medical field and journals C and F are journals in the engineering field.

Model prediction results

The data of journals, none of them appeared in the training data and the selection of journals was random in nature. Table 5 presents the results of the model predictions and the probability values predicted by the model.

As can be seen from the table, the actual prediction accuracy of the model reached 66.67%. In terms of the overall probability of prediction, the probability of predicting positive samples for both abnormal journals A and B, which are labelled as high-level warnings by the Chinese Academy of Sciences, reaches 89%, indicating the accuracy of the model's prediction. Considering the differences in indicators between disciplines, journals in the medical field with the highest number in the warning list were subsequently selected for further analysis.

Journal impact strength

Analyze abnormal and normal journals from the dimension of journal impact strength. Plotting the statistical table of impact factor, i.e., Immediacy Index and other indicators of journals AB and DE in recent years can provide a more intuitive understanding of the change in the impact of journals in recent years.

As can be seen from the statistics in the Table 6, the Immediacy Index of Journal A has been decreasing in recent years, and the index has remained at a low level.

Journal impact timeliness

From the dimension of journal impact timeliness, both early warning journals A and B have lower cited half-life than in the previous year, while normal journals have increased citation half-lives from the previous year. The cited half-life shows the number of years a journal has been cited from the current year forward to 50% of the total number of citations in the cited journals up to the current year, and measures the speed at which journals are aging (Fang, 2018). In general, the more years a journal has been cited, the more valuable

Table 5 Journal impact related indicators

Journal	Year	JCR Partition	JIF Ranking	JIF percentile	JCI Division	JCI Ranking	JCI percentile
Journal A	2021	2	52/172	70.06	1	63/329	81.00
	2020	1	28/167	83.53	1	44/315	86.19
	2019	1	29/165	82.73	1	40/312	87.34
	2018	1	32/160	80.31	1	43/304	86.02
Journal B	2021	1	18/139	87.41	1	14/195	93.08
	2020	2	59/140	58.21	2	82/188	56.65
	2019	2	64/139	54.32	3	102/187	45.72
	2018	3	99/136	27.57	3	131/185	29.46
	2017	3	86/133	35.71	3	133/185	28.38

Table 6 Journal impact strength dimension indicator data

Journal	Year	JCR Partition	Impact factor	Immediacy Index
Journal A	2022	4	1.6	0.2
	2021	4	1.515	0.208
	2020	4	1.267	0.283
	2019	4	1.198	0.247
Journal B	2021	3	2.791	0.333
	2020	N/A	N/A	N/A
Journal D	2022	2	4.20	0.8
	2021	2	4.767	0.959
	2020	2	3.88	1.096
	2019	2	3.607	0.564
Journal E	2022	1	168.9	33.7
	2021	1	202.731	47.430
	2020	1	79.323	259.056
	2019	1	60.390	15.011

it is. It can also be seen from the interpretable model given in Fig. 8 that the cited half-life plays a more important role in the final interpretation of the model predictions. The decline in the cited half-life reflects to some extent the anomalies of the journals (Table 7).

Journal features

Scholarly analyses have shown that most academic journals are published on a monthly basis and typically average about 200 articles per year (Lin et al., 2020). In contrast, most early warning journals publish significantly more than the expected annual average of 200 articles (Li et al., 2022). With the rapid expansion of journal circulation, journals often experience difficulties with manuscript processing, which reduces the average quality of papers (Lin, 2021). In terms of the number of publications, the number of publications of journal B expanded dramatically after it received its first SCI impact factor score for a special issue, with the number of publications surging by 883.6% to as many as 659 compared to 67 in the previous year, and the journal was removed from the SCI after the Chinese Academy of Sciences classified it as an early warning journal, and it was ultimately

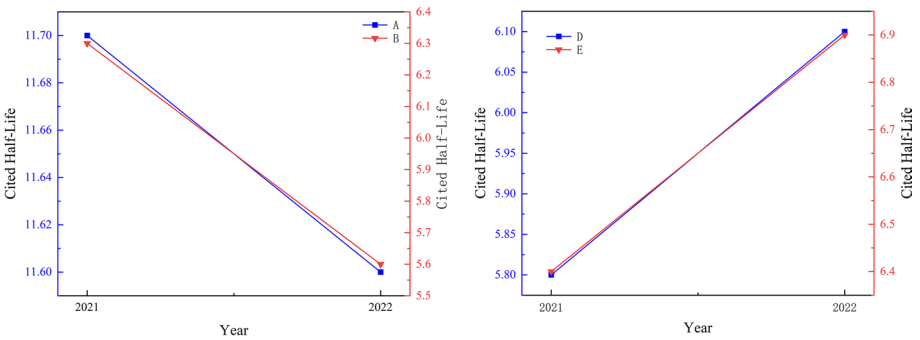


Fig. 8 Trends in cited half-life

Table 7 Journal impact timeliness dimension indicator data

Journal	Year	Cited half-life	Citing half-life
Journal A	2022	11.6	8.7
	2021	11.7	7.1
	2020	11.7	7.0
	2019	11.1	7.9
Journal B	2021	5.6	7.5
	2020	6.3	7.1
Journal D	2022	6.1	6.7
	2021	5.8	6.5
	2020	5.3	6.6
	2019	4.9	6.5
Journal E	2022	6.9	4.2
	2021	6.4	3.8
	2020	6.9	3.4
	2019	8.6	4.3

discontinued; the number of publications of journal A declined year by year, and the number of publications in 2022 dropped by twice as much as compared to that in 2019; Normal journals D and E, on the other hand, have a relatively stable number of articles per year. As shown in Fig. 9.

In addition, the dimension also involves indicators such as review cycle and contribution hit rate. According to the data we collected, the review acceptance rate of early warning journal B is very fast, spending an average of 31 days from submission to acceptance of the article; and the review cycle of the journal is about 30 days lower than that of normal

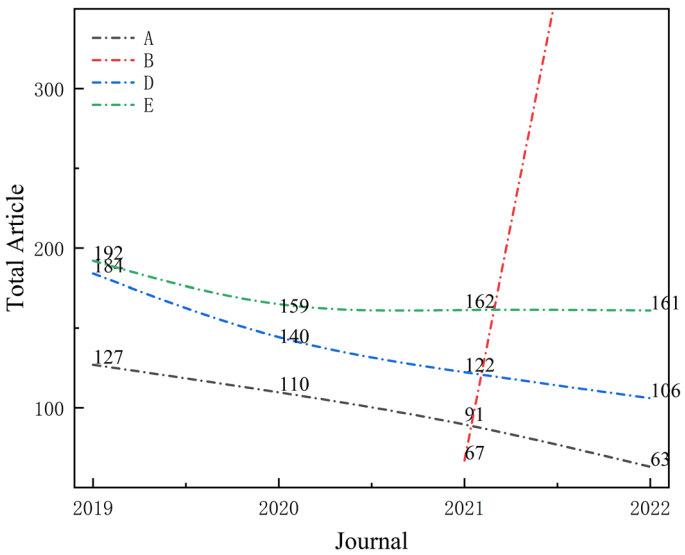


Fig. 9 Journal publications in recent years

journals. A journal's short review cycle is not a sufficient condition for judging early warning journals, but the review cycle does serve as a criterion for evaluating journal anomalies.

The self-citation rates of the four journals we selected are all relatively low, and the self-citation rate of Journal A also shows a decreasing trend year by year. It is not discussed further.

International cooperation

The POSCI has the highest feature importance in the interpretable model, suggesting that this indicator is an important factor influencing the judgment of the model. Analyzing the country of publication of the four journals, as many as 45% of the 171 articles published by Journal A in the last two years were from mainland China, and a staggering 76.52% of the 741 papers published by Journal B in the last two years were from mainland China. This abnormal value undoubtedly accelerates these two journals into the list of early warning journals. As a reference, the normal journal E has a POSCI value of 15.63%.

In summary, the analysis of the data of the combined indicators of the four journals shows that (1) the indicator of the percentage of individual countries of the anomalous journals is very high; (2) the value of the cited half-life fluctuates, which may influence the journals to be judged as anomalous journals; (3) there are anomalies in the number of annual publications and other indicators. The influence of these indicators on the results confirms the accuracy of the interpretable model in the previous section. However, there are still some shortcomings, for example, the model misclassifies normal journals D as early warning journals, which may be due to the yearly decline of the Immediacy Index indicator or the yearly decline of the number of articles. In addition, there are some other factors that we should explore, for example, we know that the abnormalities of the index of journal A may not be very obvious, and the direct cause of journal A becoming an early warning journal may be that some of its papers are produced by 'paper factories'. These external factors are not included in the indicator system and may affect the judgement of the model.

Discussion and conclusions

Conclusions

By identifying anomalous OA journals, academic reputation risks can be effectively avoided to the benefit of publishers and authors. In addition, it can assist the state in counteracting poor-quality publications and purifying the academic creative environment. This can effectively improve the quality of OA papers and prevent the loss of academic resources.

In this study, we propose an interpretable prediction model based on machine learning for identifying predatory OA journals and validate the effectiveness of the model by analysing selected medical OA journals. Specifically, we highlight the importance of paying close attention to several academic impact metrics, including the percentage of single country issuances, the volume of articles, the impact factor, and the cited half-life. Researchers must proceed with caution when multiple metrics show abnormal changes at the same time. It is also important to pay attention to metrics such as SJR, which have

widely varying values due to disciplinary differences, which will help to generalise the findings of this model to OA journals in other fields.

In addition, the analysis of early warning attribution for several existing journals through the model constructed and validated in this study revealed a 75% accuracy rate in predicting medical journals. Further analyses were conducted for medical journals. Journal A was predicted to be an early warning journal due to its high percentage of single country issuances and its characteristic of reduced citation half-life when there appeared to be no anomalies in indicators such as impact factor and self-citation rate. Similarly, Journal B was determined to be an early warning journal due to an abnormal number of publications and a disproportionately high proportion of single-country journals, as well as metrics such as review cycles. Therefore, the validity of the model is verified and can be extended. In addition, some studies have shown that the impact of early warning journals themselves is not high, with 87.69% of early warning journals having an IF value of less than 4 (Lin et al., 2020). In addition, compared with other disciplines, the scientific integrity of the biomedical field is in a worrying state, with 39.57% of papers retracted, which needs special attention (Li et al., 2020; Zhao et al., 2021).

Finally, in this paper, we constructed a risk indicator system for predatory OA journals, which currently only considers academic impact indicators due to data acquisition limitations. In the future, we will consider combining Altmetric social impact indicators to carry out more comprehensive OA journal early warning risk prediction model experiments.

Insights into the ecological construction of academic papers

The proportion of openly published papers to all papers is predicted to reach 90% by 2025, according to a study by Wolpert (2013). Despite China having fewer high-impact OA journals currently, it is the world leader in the number of OA papers produced, with a relatively higher number of papers published by Chinese authors in less reputable and less influential OA journals (Cheng & Ren, 2016). To enhance academic reputation and the construction of OA journals, and to prevent a significant outflow of research funding to early warning journals, we propose the following targeted recommendations.

(1) When submitting manuscripts, researchers should stay informed about the early warning lists in their field, as well as quality criteria such as the proportion of the percentage of single country issuances, SNIP, and self-citation rate. They should also check whether the journal has an overly fast publication cycle. Finally, they can learn about the social influence of journals through academic forums such as Medsci and Little Wood, and choose high-quality journals to avoid academic and reputational losses.
(2) Publishers should strictly control the review standards and avoid arbitrary expansion of space, which will help increase the academic impact of the journal. Furthermore, complete and transparent guidelines on publication fees, annual acceptance and rejection rates, can enhance the journal's reputation and build trust with authors.
(3) Relevant institutions should establish a quality evaluation system for OA journals that combines qualitative and quantitative methods and includes multi-faceted and comprehensive indicators, such as SNIP and Altmetric, instead of relying solely on impact factors, taking into account disciplinary differences. Early warning indicators of the academic and social impact of OA journals, including indicators of article volume, and single-country shares, should be published for the reference of researchers. Additionally, industry associations and search agencies in various subject areas are encouraged

to establish OA journal monitoring platforms that can categorize, flag, and publish abnormal OA journals within six months and provide reporting channels.

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Author contributions MKL: Collected the data, Contributed data or analysis tools, Performed the analysis, Wrote the manuscript.WJH: Comment on the overall framework of the paper, provide article revisions, and offer ideas. LTY: Collected experimental data, redid experiments, and wrote revisions. ZL: Conceived and designed the analysis, Wrote the manuscript and designed the figures, Other contribution.

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